

Homework 7: Optimization

Problem 1: Minimization using Interval Halving (25 Points)

In this problem, your goal is to find the minimum value of a function given below using two optimization methods --- one with gradient information and another using the interval halving method. Let us first start by plotting the function.

```
fun = @(x) sin(x) + sin((10.0 / 3.0) * x);  
x = linspace(-1,6,100);  
figure;  
plot(x, fun(x), '-');
```

You can see from the plot that the function has four local minima and one global minima between -1 and 6.

(a) (15 points) Write a function that can optimize any blackbox function using the interval halving method. The inputs to the function should be the lower and upper bound, the function that should be minimized and the minimum length of the interval between the upper and lower bound. The function should return the lower and the upper bound values it finds that contains the minimum of any given unimodal function.

Find the minimum value of this function between -1 and -0.5 using the interval halving function you created? You should print the lower and the upper bound values and corresponding function values at each step at those bounds. Stop when the interval between the lower and the upper bound is smaller than 0.005.

Overlay a plot of all the bounds you discovered and the corresponding function values using different markers to demonstrate how the function converges.

% Enter Your Code Here

(b) (5 points) Check the previous answer by finding the minimum value of this function between -1 and -0.5 using the "fmincon" function? Does the result depend on the starting point you specify for fmincon?

% Place your code here:

(c) (5 points) Using the "fmincon" function, find the value of x, where the value of the function "fun" is -1.2. The value of x should be between -1 and -0.5. Use a tolerance of 1e-4 for function value matching.

% Place your code here:

Problem 2: Optimization of Geometric Transformations (30 pts)

Recall the lab activity on Dilation, Rotation, and Projection

Below, we recall the lab activity on foot dilation, rotation, and projection. Here's an exercise to explore that ways matrices act on geometry

```
clearvars
close all
load('foot_3D')
% Here I remove some points that seems distracting -- they seem to be noise
--
% not really part of the foot
extra1=(foot_xyz(1,:)>0.041)&(foot_xyz(1,:)<0.075)&...
        (foot_xyz(2,:)>4.910)&(foot_xyz(2,:)<5.030)&...
        (foot_xyz(3,:)>0.480)&(foot_xyz(3,:)<0.575);
extra2=(foot_xyz(1,:)>4.925)&(foot_xyz(1,:)<5.08)&...
        (foot_xyz(2,:)>4.925)&(foot_xyz(2,:)<5.08)&...
        (foot_xyz(3,:)>0.400)&(foot_xyz(3,:)<0.871);
extra3=(foot_xyz(1,:)<0.19)|(foot_xyz(2,:)<0.19);
extra=extra1|extra2|extra3;
foot_xyz=foot_xyz(:,~extra);

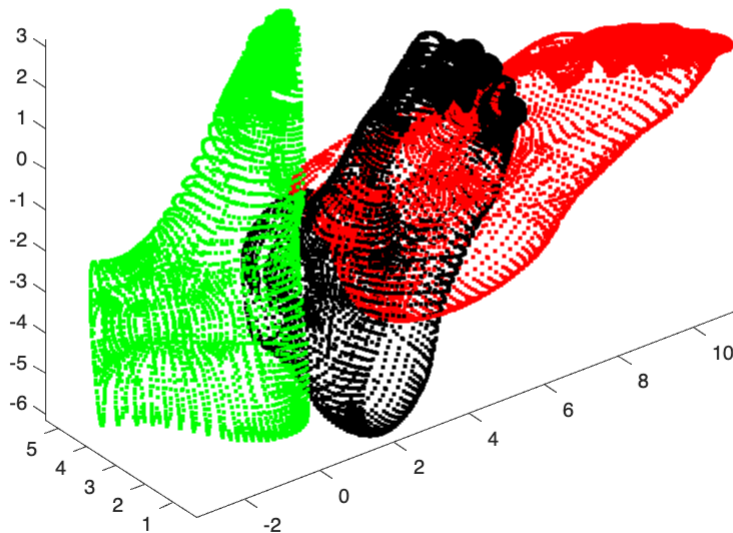
close
plot3(foot_xyz(1,:),foot_xyz(2,:),foot_xyz(3,:), 'k.')
axis equal
```

Multiply the data set `foot_xyz` by a diagonal matrix which is not the identity matrix such as $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0.6 \end{bmatrix}$. Place the result in a variable called `foot_dilation_contraction` and plot using red dots.

```
% Your code here
D=[2 0 0; 0 1 0; 0 0 0.6];
foot_dilation_contraction=D*foot_xyz;
plot3s = @(x,c) plot3(x(1,:),x(2,:),x(3,:),c);
hold on
plot3s(foot_dilation_contraction, 'r.')
```

Multiply the data set `foot_xyz` by a rotation matrix such as $\begin{bmatrix} \cos(\pi/3) & -\sin(\pi/3) & 0 \\ \sin(\pi/3) & \cos(\pi/3) & 0 \\ 0 & 0 & 1 \end{bmatrix}$. Place the result in a variable called `foot_rotation` and plot using green dots.

```
R=[cos(pi/3) -sin(pi/3) 0; sin(pi/3) cos(pi/3) 0; 0 0 1]
foot_rotation=R*foot_xyz;
plot3s(foot_rotation, 'g.')
```



The goal of this problem is to understand and apply optimization techniques in MATLAB to find optimal translation and rotation parameters that minimize the difference between a transformed and a target dataset. You will utilize matrices to perform geometric transformations such as dilation, rotation, projection, and translation on a 3D dataset representing a foot. The optimization process will involve adjusting the transformation parameters to achieve the best alignment with a given target configuration.

Dataset:

The dataset `foot_xyz` is a 3D representation of a foot. This dataset will undergo various transformations. The assignment will start with predefined transformations (dilation, rotation, projection) and will culminate in an optimization task to find the best translation and rotation parameters to match a target configuration.

a) Target Generation (5 pts)

Transform the foot by translating it by a vector $[1; 5; -2]$ and rotating it by 40 degrees around Z-axis. Your code must first apply a rotation defined by R_z to the original `foot_xyz` dataset. Then, it must apply the translation defined by `translation_vector`. Finally, it should plot both the original and the transformed datasets for comparison. The original dataset should be plotted in blue, and the transformed (target) dataset plotted in red.

% Enter Your Code Here

b) Optimization Problem (15 pts)

Objective: Find the optimal translation vector T and rotation matrix R_{opt} that minimize the difference between the transformed `foot_xyz` and a target dataset `target_xyz`.

- Use MATLAB's optimization toolbox to define and solve the optimization problem.
- The cost function should measure the Euclidean distance between the transformed dataset and the target dataset.
- Constraints: Rotation matrix must be orthogonal, and translation is bound by reasonable limits based on the dataset.

% Enter Your Code Here

c) Match a Noisy Point Cloud with Different Starting Points (10 pts)

Objective: The sensors measuring the target are corrupted. As a result, each measurement point in the target gets noisier. Let us assume that the standard deviation of noise = 0.01. Create a new target point cloud by adding a small random noise to each coordinate in all three dimensions for your original target point cloud.

Again, find the optimal translation vector T and rotation matrix R_{opt} that minimize the difference between the transformed `foot_xyz` and a target dataset `target_xyz`. Use a different initial guess this time than the last part.

- Use MATLAB's optimization toolbox to define and solve the optimization problem.
- The cost function should measure the Euclidean distance between the transformed dataset and the target dataset.
- Constraints: Rotation matrix must be orthogonal, and translation is bound by reasonable limits based on the dataset.

Find the optimal translation and rotation values.

Rotate and translate the foot and plot it so that the transformed foot is overlaid with the target point cloud.

% Enter Your Code Here

Problem 3 Maximization using Genetic algorithms (25 Points)

(a) (10 points) This assignment is designed to deepen your understanding of genetic algorithms (GAs) through hands-on experimentation and analysis. You will explore the effects of varying population sizes and mutation rates on the GA's ability to find the global maximum of a given function.

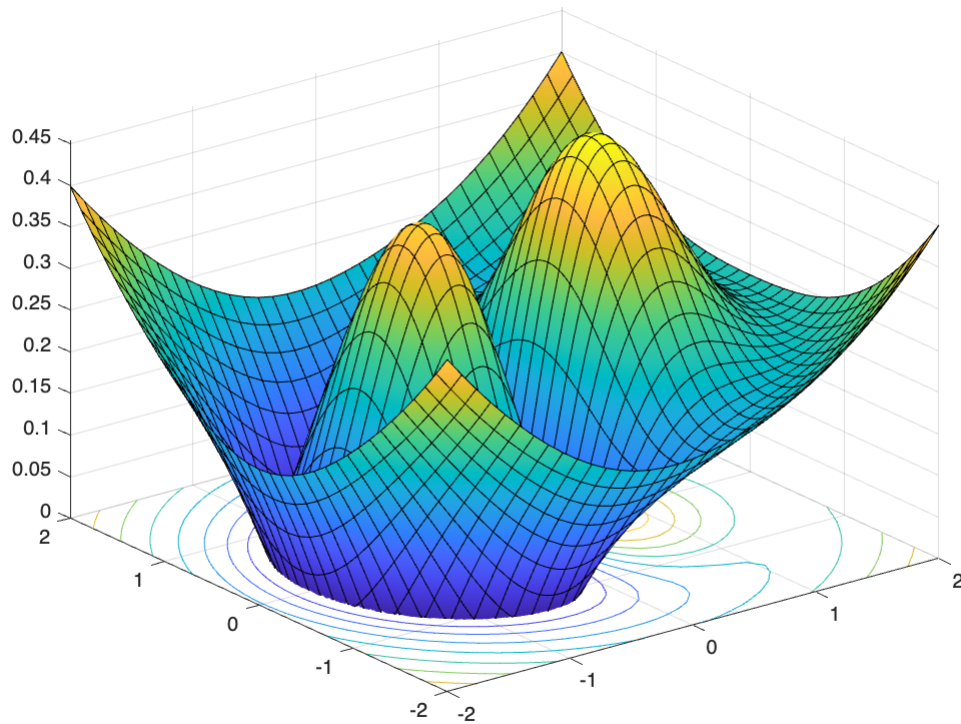
Problem Statement:

Consider the function below, where both x and y should be within the interval $[-2,2]$. Your goal is to find the maximum value of this function using a genetic algorithm.

Setup:

- Use MATLAB's ga function from the Global Optimization Toolbox.
- The function to maximize has been provided above.
- Visualize the function below

```
figure;
fsurf(@(x,y) abs(x.*exp(-x.^2-y.^2)+(x.^2+y.^2)/20),
[-2,2], 'ShowContours', 'on')
```



a) (10 points) Use a population size of 10 and run the method for a maximum of 50 generations. Report the value of x and y and objective function.

% Enter Your Code Here

b) (10 points) Run the genetic algorithm to maximize the given function with population sizes of 20, 50, and 100. Use a maximum of 50 generations for each run. Record the maximum value found, the corresponding x and y values, and the number of generations until convergence for each population size.

Your output should look like the following (Replace ?? with the actual values):

Population Size: 20, Max Value: ??, Optimal x: ??, Optimal y: ??

Population Size: 50, Max Value: ??, Optimal x: ??, Optimal y: ??

Population Size: 100, Max Value: ??, Optimal x: ??, Optimal y: ??

% Enter Your Code Here

c) (5 points) Run the genetic algorithm with mutation rates of 0.01, 0.1, and 0.2, maintaining the maximum of 50 generations. Record the maximum value found, the corresponding x and y values, and the number of generations until convergence for each mutation rate.

Your output should look like the following (Replace ?? with the actual values):

Mutation Rate: 0.01, Max Value: ??, Optimal x: ??, Optimal y: ??

Mutation Rate: 0.10, Max Value: ??, Optimal x: ??, Optimal y: ??

Mutation Rate: 0.20, Max Value: ??, Optimal x: ??, Optimal y: ??

% Enter Your Code Here

Problem 4 Optimization of a Three-Bar Truss using Genetic Algorithms (25 Points)

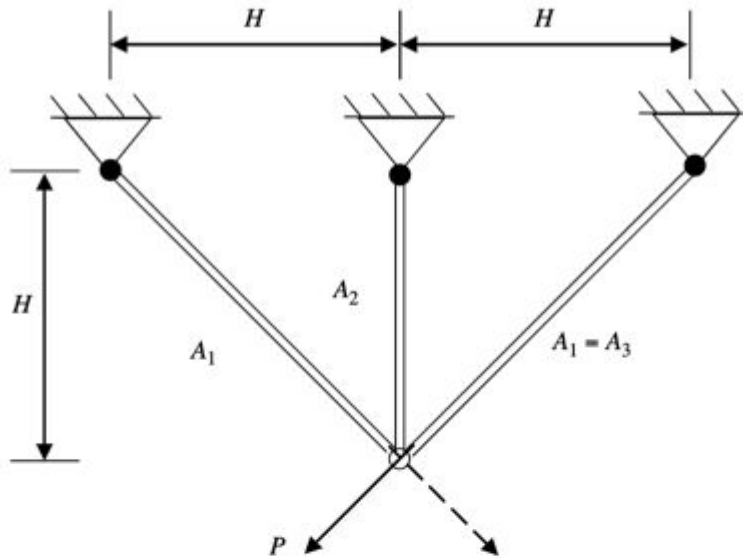
Objective:

The goal of this assignment is to apply Genetic Algorithms (GA) for the optimization of a structural engineering problem, specifically minimizing the volume of a three-bar truss subject to stress constraints. This problem introduces you to the application of GA in constrained optimization.

Problem Statement:

Consider a three-bar truss subjected to a load.

The objective is to minimize the volume of the truss while ensuring that the stress in each truss member does not exceed a specified limit. The truss is characterized by two design variables, x_1 and x_2 , representing the cross-sectional areas of the truss members. The lengths of the truss members, the applied load, and the stress limit are given constants.



given as below:

$$\text{Minimize : } f(X) = (2\sqrt{2x_1} + x_2) \times l \quad (14)$$

Subject to:

$$g_1 = \frac{\sqrt{2x_1} + x_2}{\sqrt{2x_1^2} + 2x_1x_2} P - \sigma \leq 0 \quad (15)$$

$$g_2 = \frac{x_2}{\sqrt{2x_1^2} + 2x_1x_2} P - \sigma \leq 0 \quad (16)$$

$$g_3 = \frac{1}{x_1 + \sqrt{2}x_2} P - \sigma \leq 0 \quad (17)$$

where:

$$0 \leq x_1 \leq 1 \text{ and } 0 \leq x_2 \leq 1; \quad l = 100 \text{ cm}, P = 2 \text{ KN/cm}^2, \text{ and } \sigma = 2 \text{ KN/cm}^2$$

Above image is from the following paper: Yang XS, Hossein Gandomi A (2012) Bat algorithm: a novel approach for global engineering optimization. Engineering computations 29(5):464–483 which also provides some background information about the problem

a) (10 points) Define the objective and all the constraint functions based on the problem formulation provided.

b) (15 points) Configure the GA: Set up the genetic algorithm with a population size greater than 100 and run it for 200 generations. The design variables x_1 and x_2 should be within the specified domain. Use MATLAB's ga function to find the values of x_1 and x_2 that minimize the truss volume subject to the stress constraints. Report the optimal values of x_1 and x_2 , the minimum volume achieved, and discuss the impact of the constraints on the optimization results. Hint: Experiment with different population size, generations, mutation rate, or other factors to get a final objective value less than 285. Display the optimal x_1 , x_2 , and the minimum volume.

% Enter Your Code Here

